



清华大学物理系

Department of Physics, Tsinghua University

Numerical computation of 2D generalized Brillouin zones

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Bloch band theory

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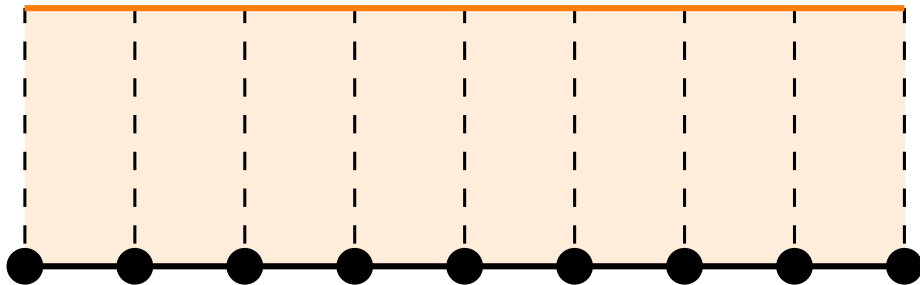
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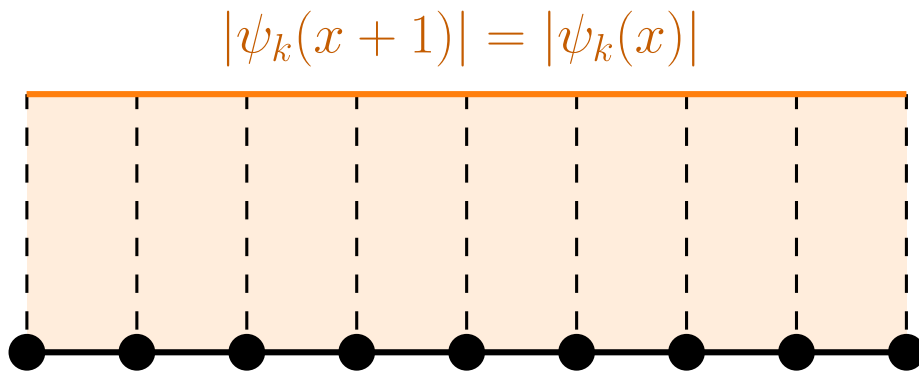
$$|\psi_k(x + 1)| = |\psi_k(x)|$$



Non-Bloch band theory

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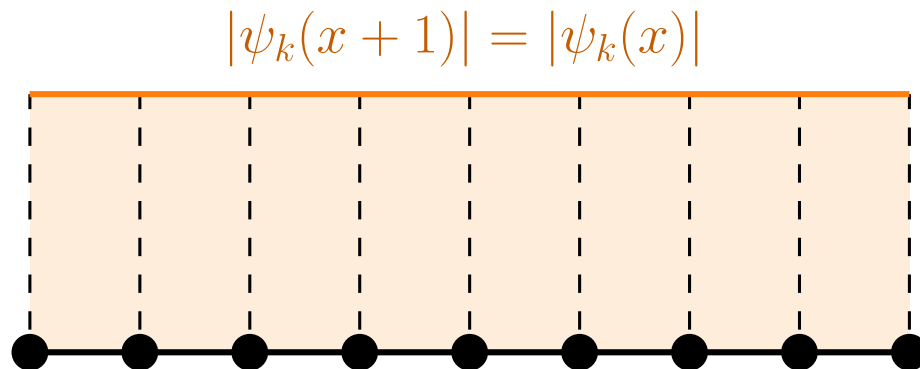


Non-Bloch band theory

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 $\psi(x + a) = \beta\psi(x), \beta \in \mathbb{C}, \beta \neq 0$

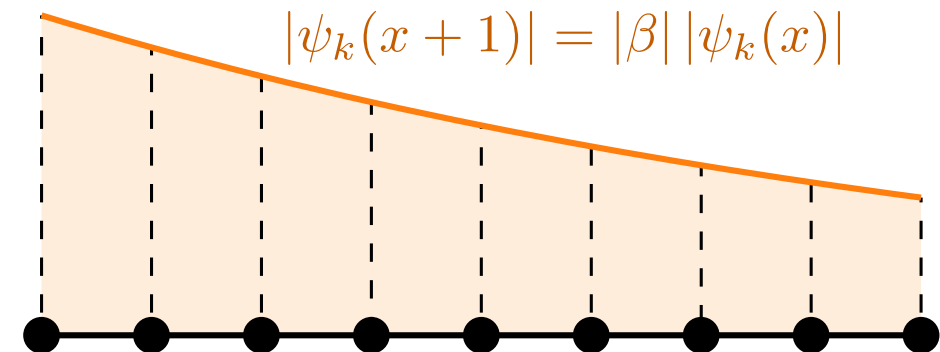
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 $\psi(x + a) = \beta\psi(x), \beta \in \mathbb{C}, \beta \neq 0$
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 Non-Bloch energy bands $E_{n(\beta)}$
- Bulk OBC states are superpositions of non-Bloch waves $\psi_{\beta,n}(x)$.



- For an n -dimensional lattice, extending $\mathbf{k} \in \mathbb{R}^n \rightarrow \beta \in \mathbb{C}^n$ **doubles the dimensionality** of momentum space.

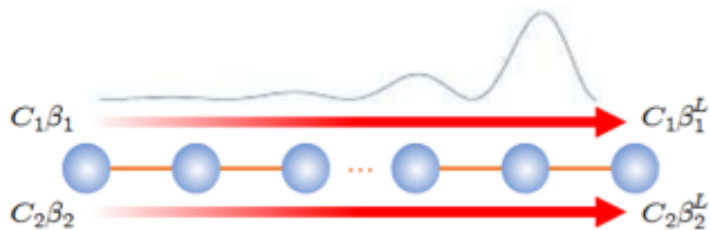
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- GBZ constraint: $|\beta^{(M)}| = |\beta^{(M+1)}|$



- Yao, S. & Wang, Z.,
Phys. Rev. Lett., 121, 086803 (2018)
- Yokomizo, K. & Murakami, S.,
Phys. Rev. Lett., 123, 066404 (2019)

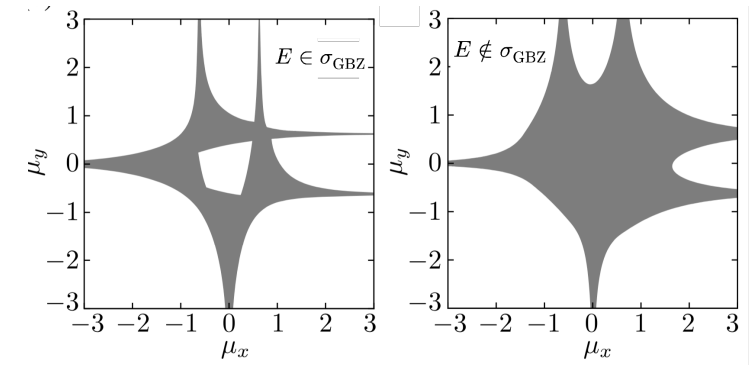
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- Amoeba formulation:

$$E \in \sigma_{\text{Amoeba}} \Leftrightarrow \text{Absence of a "central hole" in } \mathcal{A}(E)$$

- $\mathcal{A}(E) := \{\mu \in \mathbb{R}^n \mid \exists \theta, \det[E - h(e^{\mu+i\theta})] = 0\}$



Wang, H.-Y., Song, F. & Wang, Z.,
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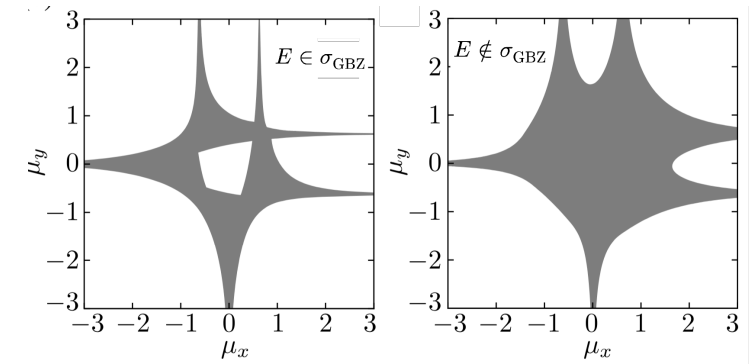
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- Ronkin function:

$$R_E(\boldsymbol{\mu}) = \int_{\mathbb{T}^n} \left(\frac{d\boldsymbol{\theta}}{2\pi} \right)^n \ln |\det[E - h(e^{\boldsymbol{\mu} + i\boldsymbol{\theta}})]|$$

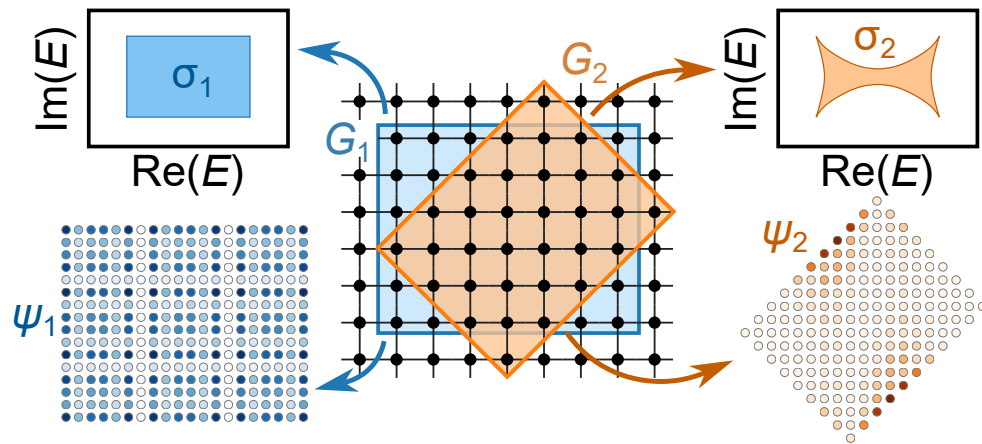
- Central hole: an open complement component where $\nabla R_E = 0$.



Wang, H.-Y., Song, F. & Wang, Z.,
Phys. Rev. X, 14, 021011 (2024)

Geometry-dependent skin effect

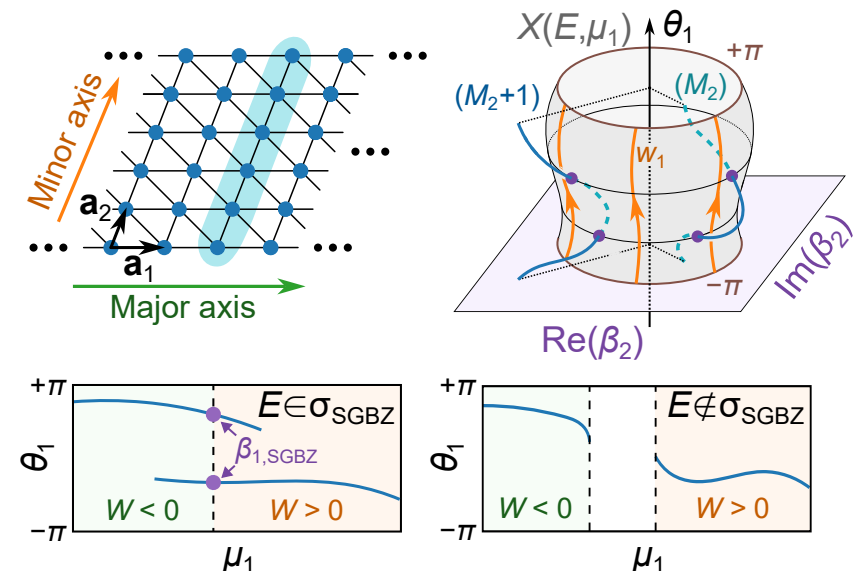
- In $n \geq 2$ dimensions, the OBC spectrum and decay factors depend on boundary geometry.



Zhang, K., et al., *Nat. Commun.*, 13, 2496 (2022)

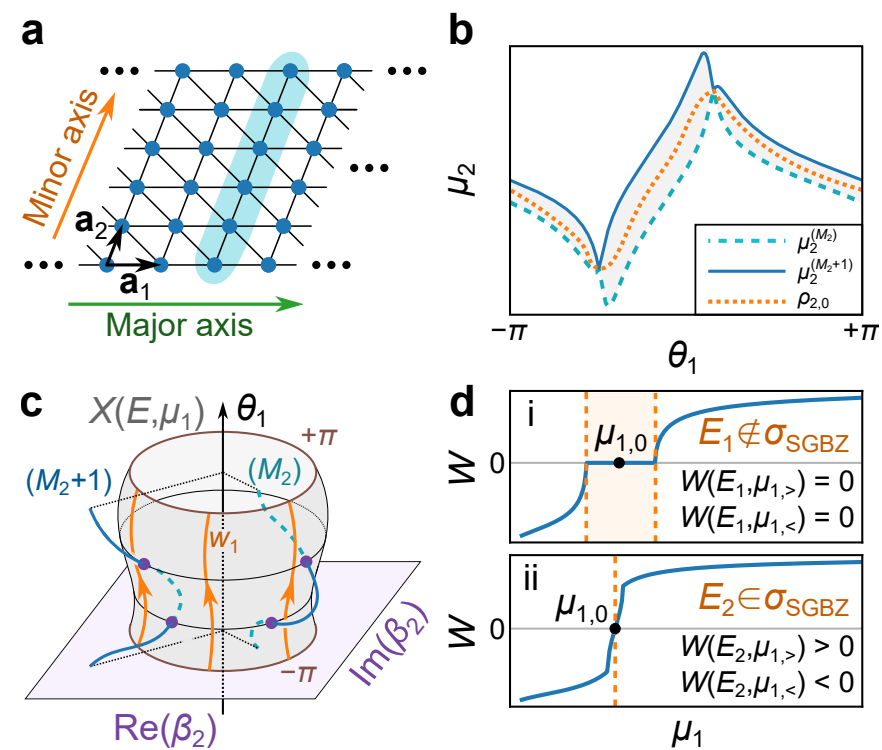
Strip-GBZ formulation

- OBC spectrum in the strip limit ($L_1 \gg L_2$).



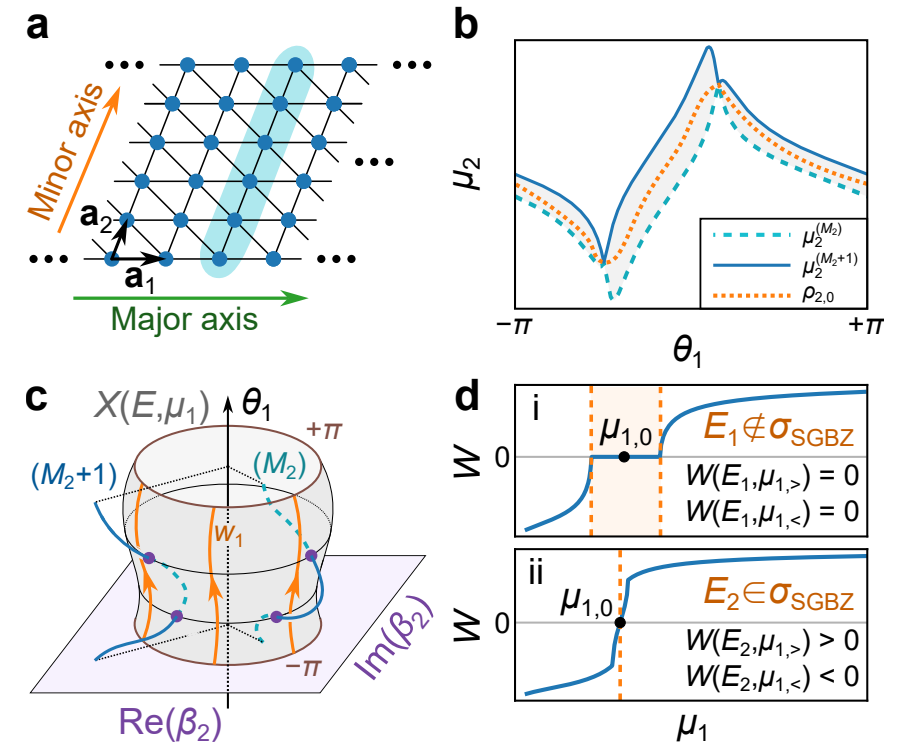
Xiong, Y., et al., *arXiv*, 2407.01296 (2024)

Wang, C., et al., *arXiv*, 2506.22743 (2025)



- Minor-axis log-radius function: $\rho_{2,0}(\theta_1; E, \mu_1)$

Periodic, with $\mu_2^{(M_2)} \leq \rho_{2,0} \leq \mu_2^{(M_2+1)}$



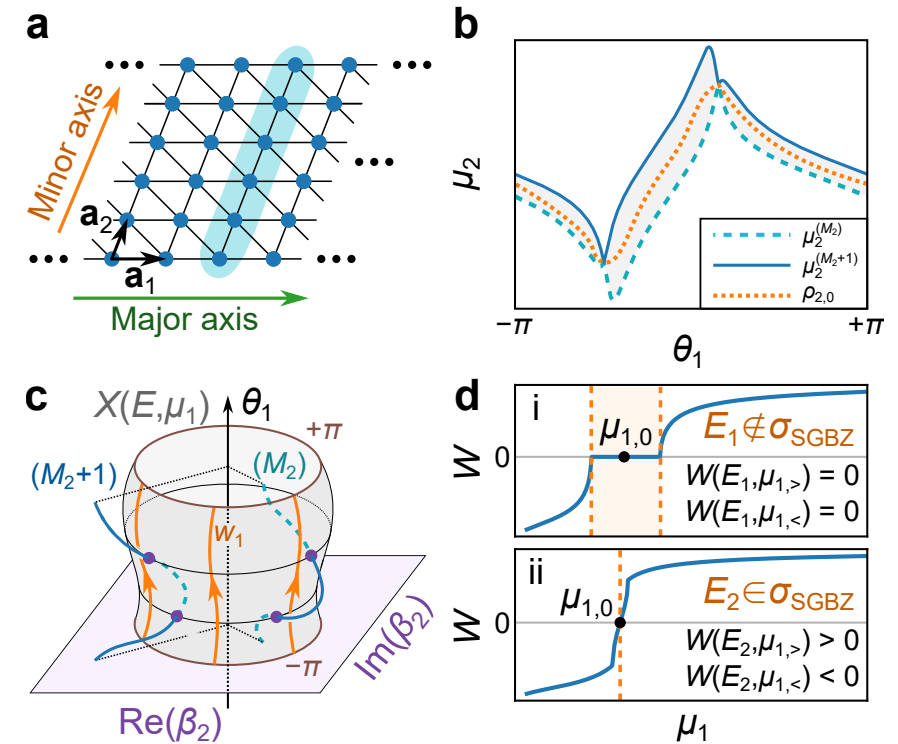
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Periodic, with $\mu_2^{(M_2)} \leq \rho_{2,0} \leq \mu_2^{(M_2+1)}$

- Base manifold:

$$X(E, \mu_1) := \{(\beta_1, \beta_2) \mid \theta_1, \theta_2 \in [-\pi, \pi]$$

$$\beta_1 = e^{\mu_1 + i\theta_1}, \beta_2 = e^{\rho_{2,0}(\theta_1) + i\theta_2}\}$$



- Minor-axis log-radius function: $\rho_{2,0}(\theta_1; E, \mu_1)$

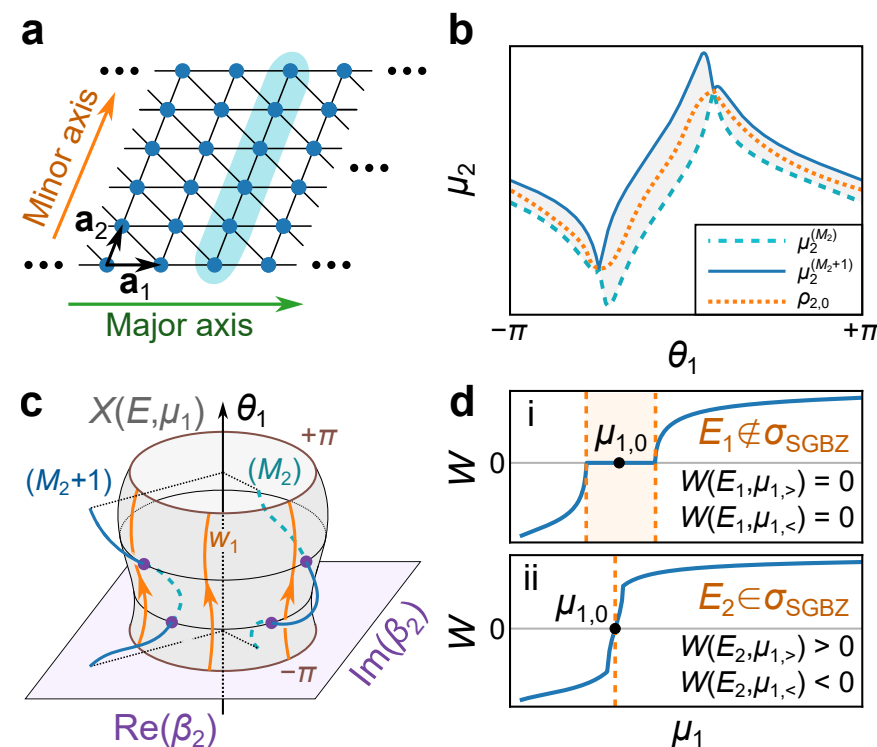
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- Major-axis winding number $w_1(\theta_2; E, \mu_1)$:
Winding of $\det(E - h)$ along θ_1 at fixed θ_2 .



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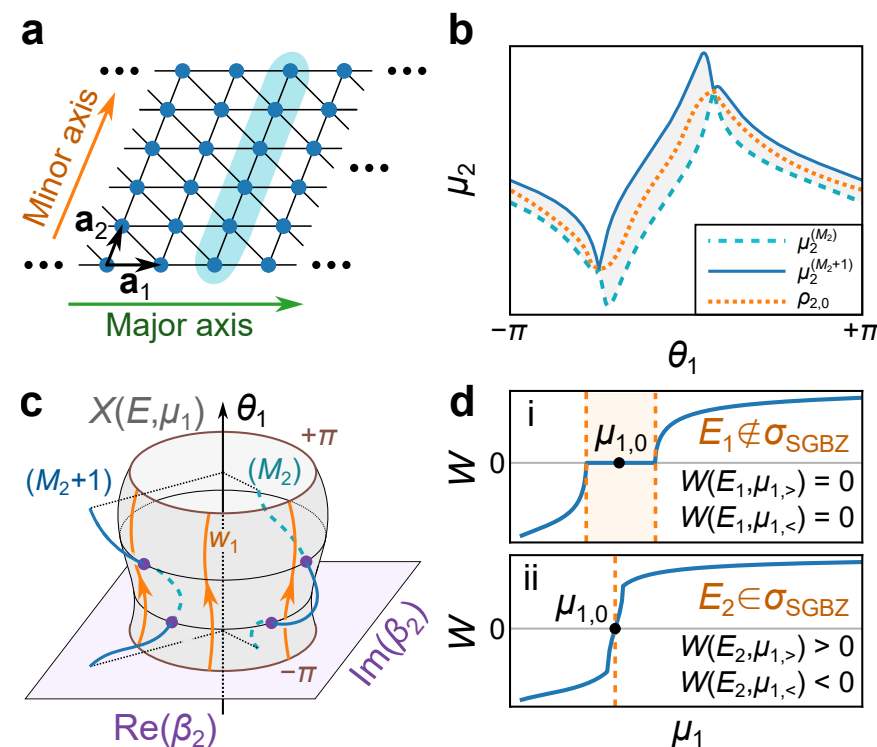
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- Major-axis winding number $w_1(\theta_2; E, \mu_1)$:
Winding of $\det(E - h)$ along θ_1 at fixed θ_2 .

- Average winding number:

$$W(E, \mu_1) := \frac{1}{2\pi} \int_{-\pi}^{\pi} w_1(\theta_2; E, \mu_1) d\theta_2$$

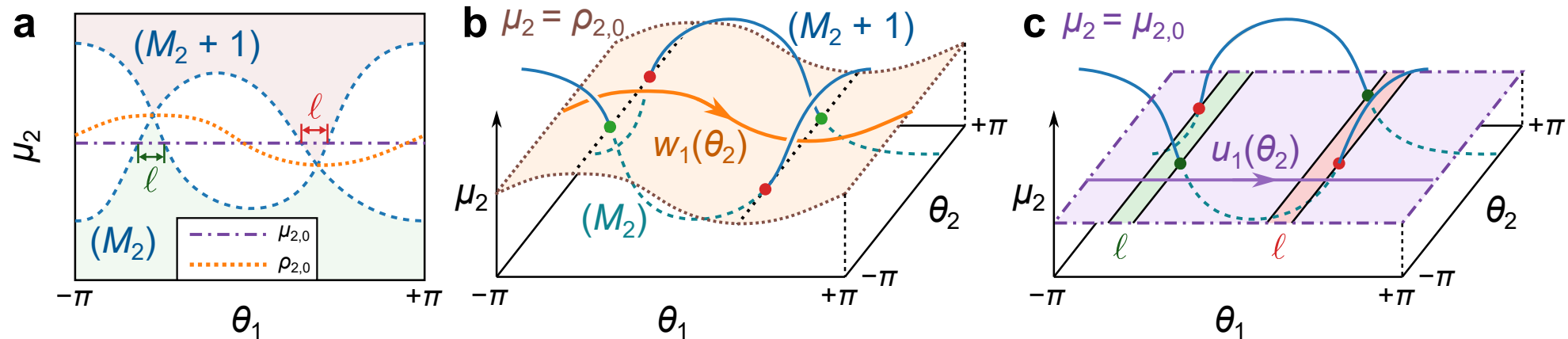


- Derivative of $R_E(\mu) \Leftrightarrow$ average winding number on the torus at fixed μ

$$\frac{\partial R_E(\mu)}{\partial \mu_j} = \int_{\mathbb{T}^{n-1}} \frac{d\theta_1 \cdots d\hat{\theta}_j \cdots d\theta_n}{(2\pi)^{n-1}} u_j(\theta_1, \dots, \hat{\theta}_j, \dots, \theta_n)$$

where

$$u_j(\theta_1, \dots, \hat{\theta}_j, \dots, \theta_n) := \int_0^{2\pi} \frac{d\theta_j}{2\pi i} \frac{\partial_{\theta_j} \det(E - h(e^{\mu+i\theta}))}{\det(E - h(e^{\mu+i\theta}))}$$



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- $\det(E - h(\beta))$: a **continuous** map from the base manifold $\beta \in X$ to the complex plane.

- Projection onto the unit circle: $P : \mathbb{C} \setminus \{0\} \rightarrow S^1, z \mapsto z/|z|$

- Composition:

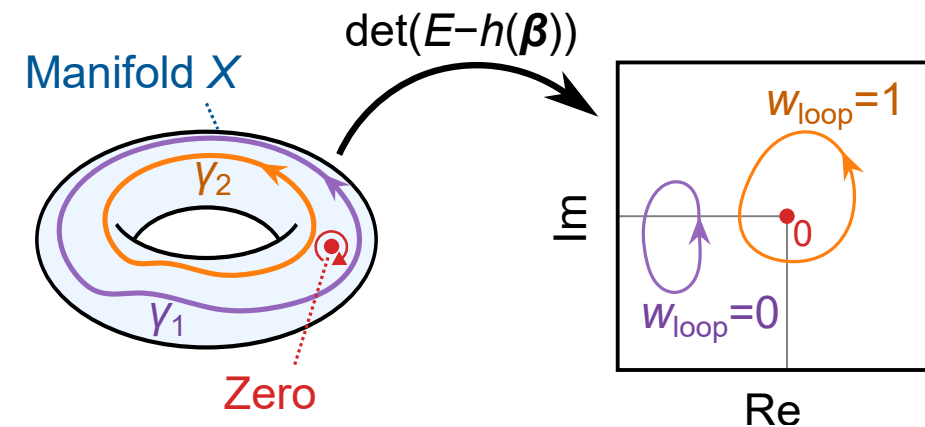
$$X \setminus \{\text{Zeros}\} \xrightarrow{\det[E-h(\beta)]} \mathbb{C} \setminus \{0\} \xrightarrow{P} S^1$$

- Winding number: homomorphism

$$w_{\text{loop}} : \pi_1(X \setminus \{\text{Zeros}\}) \rightarrow \pi_1(S^1) = \mathbb{Z}$$

- Since $\mathbb{Z}$ is Abelian,

$$w_{\text{loop}} : H_1(X \setminus \{\text{Zeros}\}) \rightarrow \mathbb{Z}$$



- Strip GBZ: $X = X(E, \mu_1)$
- Amoebic GBZ: $X = X(\mu) := \{(e^{\mu_1 + i\theta_1}, e^{\mu_2 + i\theta_2}) \mid \theta_1, \theta_2 \in [-\pi, \pi]\}$

- Argument principle:

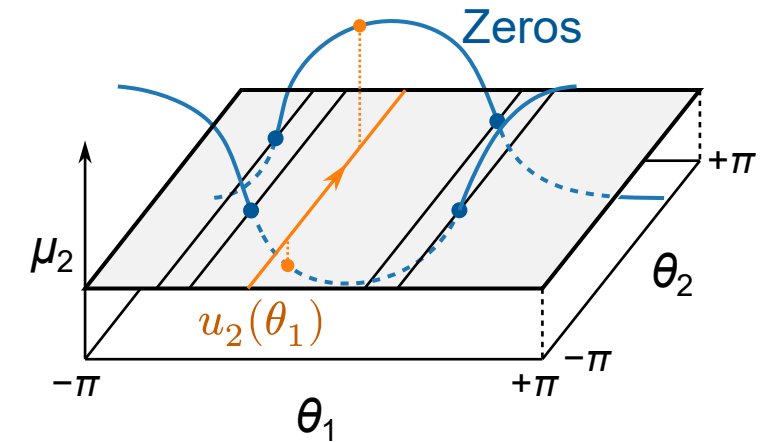
$$\oint_{\mathcal{C}} \frac{dz}{2\pi i} \frac{f'(z)}{f(z)} = N_{\text{zeros}} - N_{\text{poles}}$$

Zeros and poles inside $\mathcal{C}$ count with multiplicity. The contour avoids both.

- Let $-M_j$ and N_j be the lowest and highest powers of β_j in $\det(E - h)$, for $j = 1, 2$.
- Loop winding number:
 - $u_1(\theta_2; \mu_1, \mu_2) = (\# \text{ of } \beta_1 \text{ roots inside } e^{\mu_1}) - M_1$
 - $u_2(\theta_1; \mu_1, \mu_2) = (\# \text{ of } \beta_2 \text{ roots inside } e^{\mu_2}) - M_2$

Algorithm: Compute $\partial_2 R_E(\mu_1, \mu_2)$

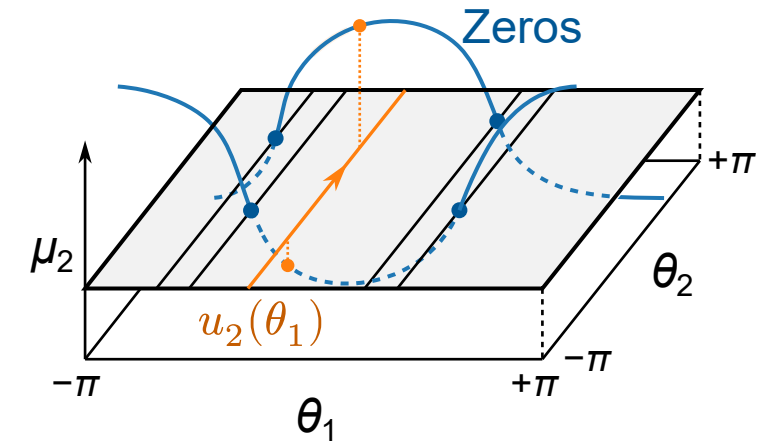
- 1 Compute tracks of roots $(E, \mu_1, \theta_1) \mapsto \beta_2^{(j)}$
 - 2 Collect the intersection points of $\ln|\beta_2^{(j)}| = \mu_2$
 - 3 **if** intersection points are discrete:
 - 4 Partition the θ_1 circle at the crossing angles
 - 5 **For each** interval:
 - 6 Choose the interval midpoint θ_1
 - 7 Count roots with $\ln|\beta_2| < \mu_2$: $u_2 = N - M_2$
 - 8 $\partial_2 R_E = \sum_{\text{intervals}} u_2 \times (\text{interval width})/2\pi$
 - 9 **else**: $\partial_2 R_E$ is ill-defined.
-



- $u_1(\theta_2; \mu_1, \mu_2) =$
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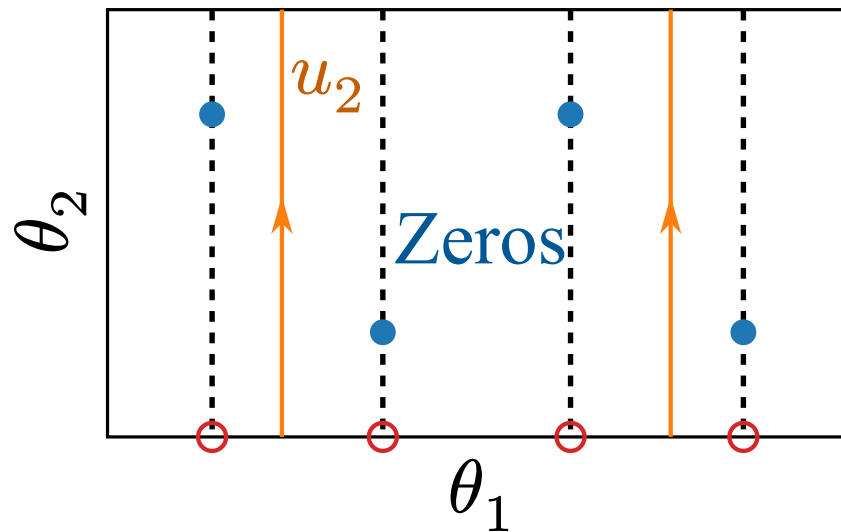
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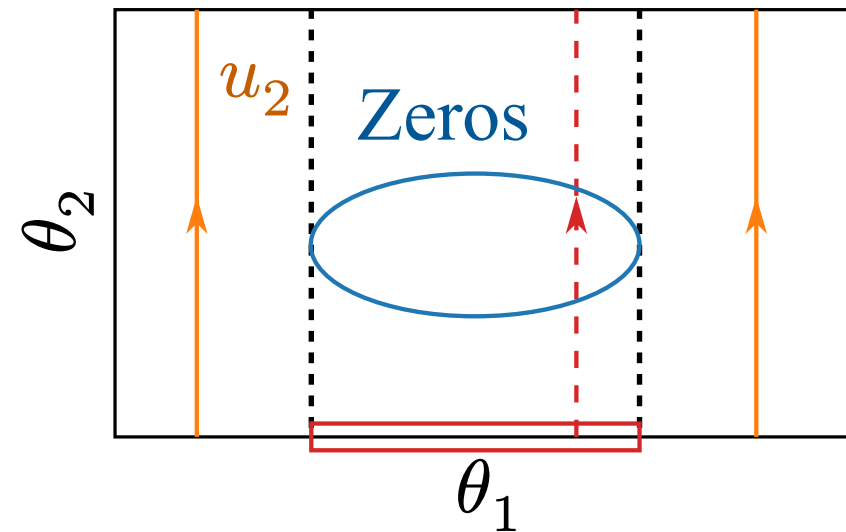
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- $u_2(\theta_1; \mu_1, \mu_2) =$
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- $\partial_1 R_E$ can be computed
 with the same procedure.

Discrete zeros (point subsets)



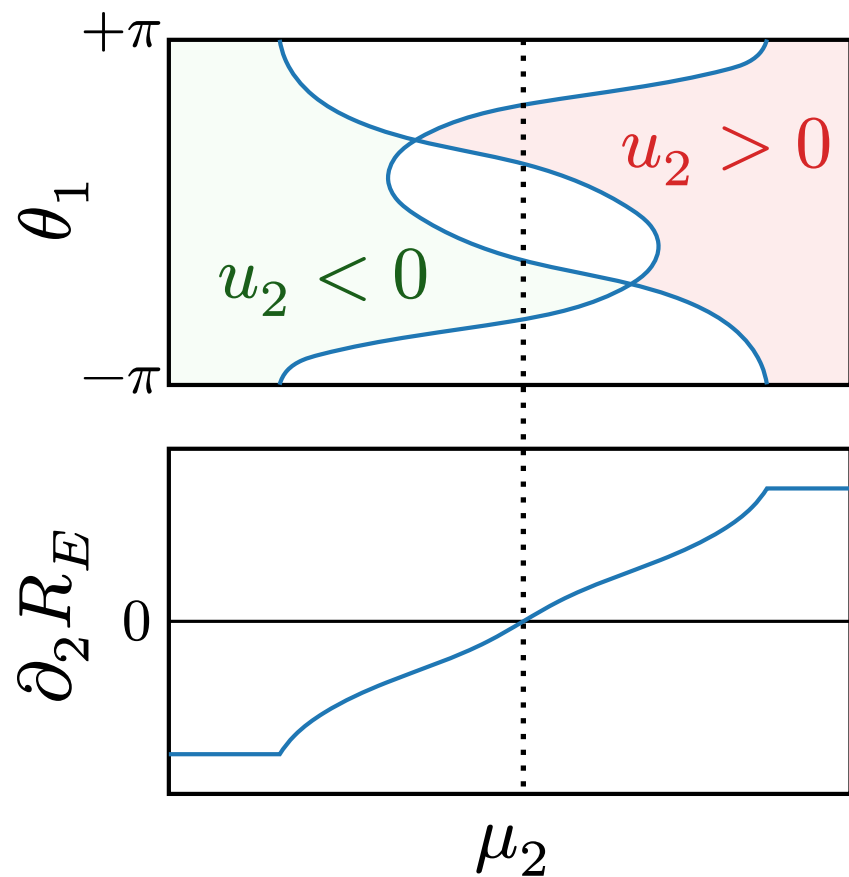
- Singular loop parameters form a set of measure zero.
- $\int_{-\pi}^{\pi} u_2(\theta_1) d\theta_1$ is **well defined**.

Continuum zeros (line subsets)

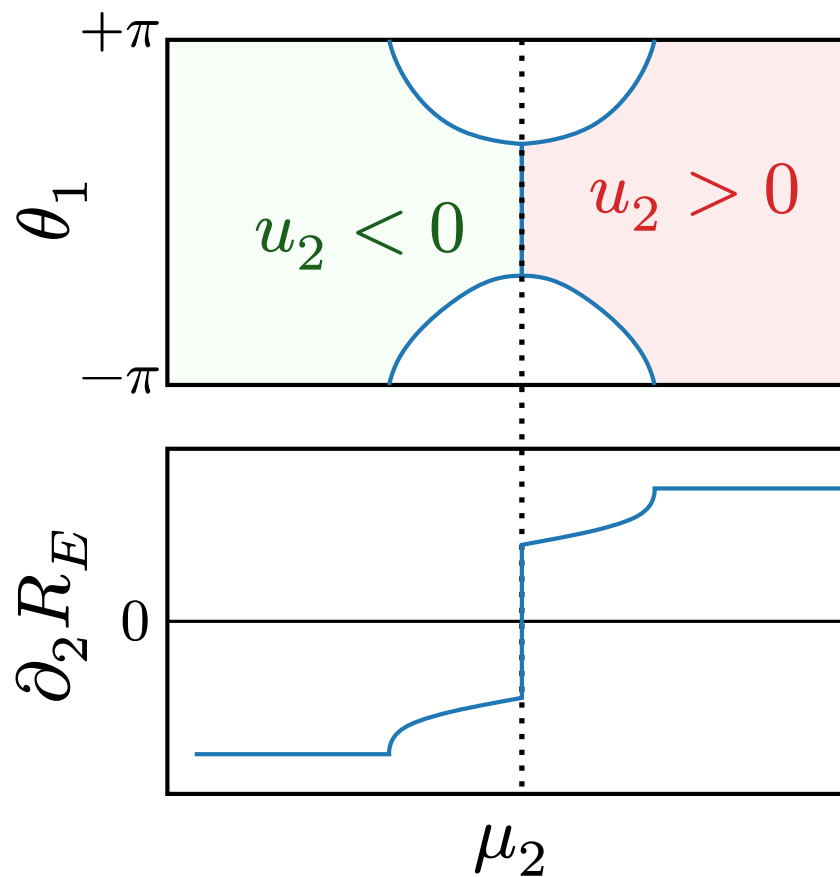


- Singular loop parameters occupy an interval.
- $\int_{-\pi}^{\pi} u_2(\theta_1) d\theta_1$ is **ill-defined**.

Discrete zeros (point subsets)



Continuum zeros (line subsets)



- At fixed E , GBZ subsets satisfy $f(E, \beta_1, \beta_2) = 0$ and $\ln|\beta| = \mu_{\min}$.
- Partial minimization preserves convexity: $F_E(\mu_1) := \min_{\mu_2} R_E(\mu_1, \mu_2)$ is convex.

Outer search in μ_1

- 1 Bracket the sign change in $\partial_1 R_E$.
 - 2 Choose the midpoint μ_1 .
 - 3 Track $\beta_2^{(j)}(\theta_1)$ once.
 - 4 Run the inner search for μ_2 .
 - 5 Evaluate $\partial_1 R_E$ at that result.
 - 6 Update the μ_1 bracket.
 - 7 Stop at zero winding or a resolved continuum minimum.
-

Inner search in μ_2

- 1 Reuse the tracks at fixed (E, μ_1) .
 - 2 Test the gap and flat-track levels.
 - 3 Bracket the sign change in $\partial_2 R_E$.
 - 4 Locate crossings of $\ln|\beta_2| = \mu_2$.
 - 5 Evaluate $\partial_2 R_E$ by root counts.
 - 6 Bisect until $|\partial_2 R_E| < \varepsilon_w$.
 - 7 Retain refined roots for later levels.
-

- A flat root track at $\ln|\beta_2| = \mu_{2,c}$ makes the usual loop winding singular over an interval of θ_1 .

- Probe the two sides of the level:

$$w_2^- = \partial_2 R_E(\mu_1, \mu_{2,c} - \varepsilon), \quad w_2^+ = \partial_2 R_E(\mu_1, \mu_{2,c} + \varepsilon).$$

- If $w_2^- < 0 < w_2^+$, the level minimizes the convex function in μ_2 . The minimum can occur at a kink where an ordinary derivative does not exist.
- Resolve the outer search with analogous μ_1 probes at fixed $\mu_{2,c}$. Opposite signs identify a continuum minimum and produce line subsets.
- If the signs do not straddle zero, continue on the appropriate side. A zero-winding plateau with no zeros represents an exterior energy.

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- Fix (E, μ_1) and vary θ_1 , with $\beta_1 = \exp(\mu_1 + i\theta_1)$. Solve $f(E, \beta_1, \beta_2) = 0$ for all $K = M_2 + N_2$ roots.
- A uniform θ_1 mesh can underresolve rapid root motion near a multiple root. Sorting roots independently at each step also loses their track identities.
- The continuation state is

$$\mathbf{y} = \left(\theta_1, \ln \beta_2^{(1)}, \dots, \ln \beta_2^{(K)} \right).$$

- Adapt the angular step to motion in this state space. At each new angle, solve the polynomial again and match the roots to their predicted positions.

Implicit differentiation along a regular root gives

$$\partial_{\beta_1} f \cdot i\beta_1 + \partial_{\beta_2} f \cdot \frac{d\beta_2}{d\theta_1} = 0.$$

Hence the logarithmic tangent of track j is

$$V_j := \frac{d \ln \beta_2^{(j)}}{d\theta_1} = -i \frac{\beta_1}{\beta_2^{(j)}} \frac{\partial_{\beta_1} f}{\partial_{\beta_2} f}, \quad \mathbf{V} = (1, V_1, \dots, V_K).$$

The proposed angular increment is

$$\Delta\theta_1 = \frac{h}{\|\mathbf{V}\|_2}, \quad \|\mathbf{V}\|_2 = \sqrt{1 + \sum_j |V_j|^2}.$$

- Fast root motion increases $\|\mathbf{V}\|_2$ and reduces $\Delta\theta_1$.
- h controls the approximate arclength step and adapts to prediction error.
- Undefined tangents of zero/infinite padding roots are excluded from the norm.

1 **Predict:** $\hat{\beta}_2^{(j)} = \beta_2^{(j)} \exp(V_j \Delta \theta_1)$.

2 **Solve:** compute all roots $\tilde{\beta}_2^{(k)}$ at the trial angle.

Match: find a one-to-one assignment π minimizing

3
$$\sum_j d_{\text{chord}} \left(\hat{\beta}_2^{(j)}, \tilde{\beta}_2^{(\pi(j))} \right).$$

4 **Accept or retry:** compare matched roots with the predictions.

- Hungarian matching uses distance on the Riemann sphere:

$$d_{\text{chord}}(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}.$$

- The sphere metric handles roots near zero and infinity. A prediction anchor reduces track swaps at close approaches.
- Each regular root sample comes from a polynomial solve.

Use the largest matched prediction error:

$$e = \frac{\max_j d_{\text{chord}}\left(\hat{\beta}_2^{(j)}, \tilde{\beta}_2^{(\pi(j))}\right)}{\text{atol} + \text{rtol} \cdot \text{median}_{k:\text{finite}} |\tilde{\beta}_2^{(k)}|}.$$

Accepted step: $e < 1$

- Store the solved roots in track order.
- Propose $h_{\text{next}} = h \min\left(10, 0.9e^{-\frac{1}{2}}\right)$.
- For $e = 0$, use the maximum growth factor.

Rejected step: $e \geq 1$

- Reduce h and solve again.
- Use $h \leftarrow h \max\left(0.2, 0.9e^{-\frac{1}{2}}\right)$.
- Stop after the retry budget or minimum step is reached.

Defaults: $\text{atol} = 10^{-12}$, $\text{rtol} = 10^{-3}$, $h_{\text{max}} = 0.5$. An accepted step cannot grow immediately after a rejection.

Trigger	Candidate condition
Step collapse	$\Delta\theta_1 < 10^{-10}$ near a singular implicit derivative.
Pairwise approach	For a nearby root pair, $d \beta_i - \beta_j ^2 / d\theta_1$ changes from negative to positive.

- Refine a point candidate by solving $f = 0$ and $\partial_{\beta_2} f = 0$. For an interval candidate, first locate its distance-derivative zero.
- A close approach is only a candidate. Confirm a finite-root cluster before splitting the segment.
- Store the multiple root as a shared endpoint, restart beyond it, and track the next segment.
- At $\theta_1 = 2\pi$, retain the permutation relating the closing roots to $\theta_1 = 0$.

- ZeroManager returns tracked roots, analytic tangents, and multiple-root boundaries on an adaptive mesh.
- **SGBZ**: refine equal-modulus events, build $\mu_{2,\text{mid}}(\theta_1)$, and extract boundary points or line intervals.
- **Amoeba**: locate crossings of $\ln|\beta_2| = \mu_2$ and reuse the same tracks throughout the inner μ_2 search.
- Cubic Hermite interpolation supplies matching anchors and the SGBZ loop path. Singular endpoint derivatives require linear fallback.
- Step control resolves root motion. Additional crossing and extremum refinement resolves the level sets used by each solver.

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- Introduction to 2D generalized Brillouin zones
- Computing average winding via polynomial root-solving
- Pseudo-arclength continuation of zeros
- Benchmark: 2D complex Hatano-Nelson model



$$h(\beta_x, \beta_y) = J_{x1}\beta_x^{-1} + J_{x2}\beta_x + J_{y1}\beta_y^{-1} + J_{y2}\beta_y.$$

- Factorization: $J_{\alpha 1} = e^{\gamma_{\alpha} + i\delta_{\alpha}} J_{\alpha}$, $J_{\alpha 2} = e^{-\gamma_{\alpha} + i\delta_{\alpha}} J_{\alpha}^*$, $\alpha = x, y$
 - The Amoeba GBZ, x -strip GBZ and y -strip GBZ are identical.
- a**

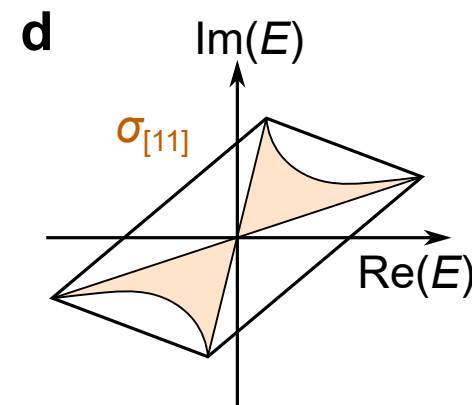
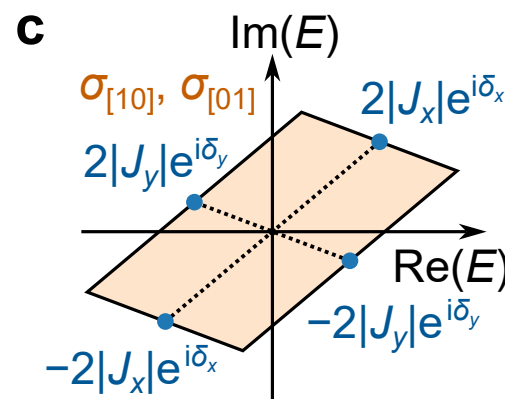
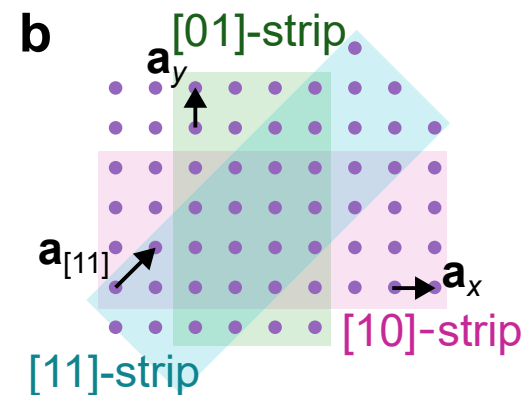
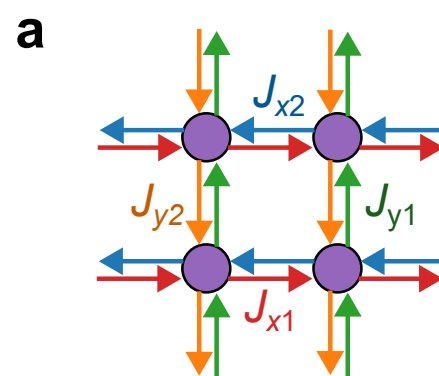
$$\beta_x = \exp(\gamma_x + \mathrm{i}\theta_x),$$

$$\beta_y = \exp(\gamma_y + \mathrm{i}\theta_y),$$

- The $[11]$ -strip GBZ:

$$\tilde{\beta}_{[11]} = e^{\gamma_x + \gamma_y + i\theta_{[11]}},$$

$$\tilde{\beta}_y = e^{\gamma_y + i\theta_y} \sqrt{\left| \frac{J_x^* e^{i\Delta_{xy} + i\theta_{[11]}} + J_y}{J_x e^{i\Delta_{xy} - i\theta_{[11]}} + J_y^*} \right|},$$



For every returned point and every stored line sample, compare

$$\varepsilon_{\mu} = \max_{j=1,2} \max_n |\ln|\beta_{j,n}| - \mu_j^{\text{exact}(\theta_{1,n})}|,$$

$$\varepsilon_E = \max_n \frac{|E - h(\beta_{x,n}, \beta_{y,n})|}{|E| + \left|\frac{J_{x1}}{\beta_{x,n}}\right| + |J_{x2}\beta_{x,n}| + \left|\frac{J_{y1}}{\beta_{y,n}}\right| + |J_{y2}\beta_{y,n}|}.$$

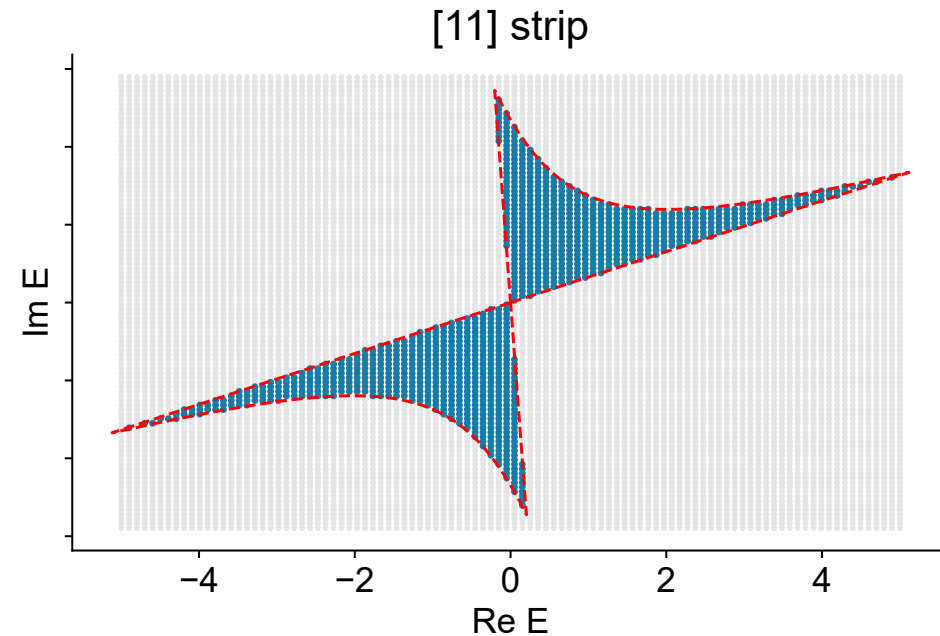
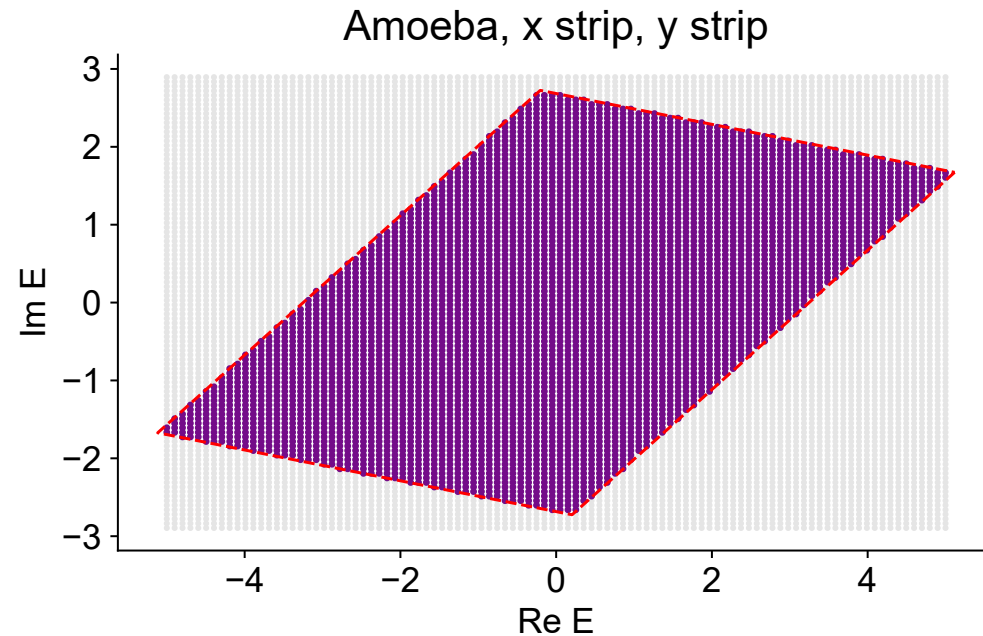
Case	$(J_{x1}, J_{x2}, J_{y1}, J_{y2})$	E
Point subsets	$(1 + i, 1.5 + 1.2i, -1 + i, -1.2 - 0.5i)$	$1 + i$
Line subsets	$(1, 1.5, -1, -1.2)$	1

- Acceptance thresholds: $\varepsilon_{\mu} \leq 10^{-5}$ and $\varepsilon_E \leq 10^{-7}$.
- Check spectral membership independently, including successful empty results.

Case	GBZ	Samples	ϵ_{μ}	ϵ_E
Points	Amoeba	4	9.52×10^{-9}	2.01×10^{-16}
Points	x strip	4	2.00×10^{-9}	1.00×10^{-16}
Points	y strip	4	1.44×10^{-9}	1.00×10^{-16}
Points	[11] strip	4	1.44×10^{-9}	2.10×10^{-16}
Lines	Amoeba	354	5.97×10^{-7}	1.32×10^{-10}
Lines	x strip	350	4.22×10^{-7}	1.32×10^{-10}
Lines	y strip	346	5.80×10^{-7}	1.28×10^{-10}
Lines	[11] strip	380	3.98×10^{-7}	5.51×10^{-11}

All eight cases pass. Each point case contains four points and each line case contains two line subsets.

NumPy backend, default solver settings.



- All four methods evaluate the same 100×100 grid, with zero failed results.
- Amoeba, x strip, and y strip share all 4,832 in-spectrum grid points. The [11] strip contains 1,108 points on the same grid.

Dots are computed grid points; red dotted lines are the closed-form GBZ boundaries.